1	2	3	4	5	6	Total
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Name: ______________________________ May 12, 2019 Sunday

Department/School:____________________ SID:__________________________________

University Physics I

Midterm Examination

Kuang Yaming Honors School, Nanjing University

<u>Select five out of following six problems.</u>

1. (20pts.) A of mass m moves along the positive x-axis ($x \ge 0$) with an attractive inverse-cube force field $F = -k/x^3$ acting on it, where k is a positive constant. The particle starts from rest at $x_0\ (>0)$.

(a) Find the velocity of the particle as a function of coordinate x.

(b) Find the velocity of the particle as a function of time t.

(c) Show that the time taken for the particle to reach the origin is $T_0 = x_0^2\sqrt{m/k}$.

Solution: (a) The equation of motion of the particle is

$$m\frac{dv}{dt} = -\frac{k}{x^3}.$$

Using $\dfrac{dv}{dt} = \dfrac{dv}{dx}\dfrac{dx}{dt} = v\dfrac{dv}{dx}$, the above equation can be rewritten as

$$mv\frac{dv}{dx} = -\frac{k}{x^3} \quad \text{or} \quad vdv = -\frac{k}{m}\frac{dx}{x^3}.$$

Integrating both sides of above equation, we have

$$\int_0^v vdv = -\frac{k}{m}\int_{x_0}^x \frac{dx}{x^3} \quad \text{or} \quad \frac{1}{2}v^2 = \frac{k}{2m}\left(\frac{1}{x^2} - \frac{1}{x_0^2}\right).$$

Thus

$$v(x) = \sqrt{\frac{k}{m}\left(\frac{1}{x^2} - \frac{1}{x_0^2}\right)} = \sqrt{\frac{k}{m}\frac{x_0^2 - x^2}{x^2x_0^2}} = \frac{1}{x_0}\sqrt{\frac{k}{m}}\frac{\sqrt{x_0^2 - x^2}}{x}.$$

(b) From

$$\frac{dx}{dt} = v = \frac{1}{x_0}\sqrt{\frac{k}{m}}\frac{\sqrt{x_0^2 - x^2}}{x},$$

we obtain

$$\frac{xdx}{\sqrt{x_0^2 - x^2}} = \frac{1}{x_0}\sqrt{\frac{k}{m}}dt.$$

Integrating both sides of equation, we have

$$\int_{x_0}^x \frac{xdx}{\sqrt{x_0^2 - x^2}} = \frac{1}{x_0}\sqrt{\frac{k}{m}}\int_0^t dt \quad \text{or} \quad \sqrt{x_0^2 - x^2} = \frac{1}{x_0}\sqrt{\frac{k}{m}}t.$$

Therefore

$$x^2 = x_0^2 - \frac{k}{mx_0^2}t^2 \quad \text{or} \quad x = x_0\sqrt{1 - \frac{k}{mx_0^4}t^2}. \tag{1}$$

Making derivative of above equation to time t, we obtain

$$v(t) = \frac{dx}{dt} = x_0\frac{1}{2}\left(1 - \frac{k}{mx_0^4}t^2\right)^{-\frac{1}{2}}\left(-\frac{k}{mx_0^4}2t\right) = -\frac{k}{mx_0^3}\frac{t}{\sqrt{1 - \dfrac{k}{mx_0^4}t^2}}.$$

(c) Substituting $t = T_0$, $x = 0$ into Eq. (1), we have

$$0 = x_0 \sqrt{1 - \frac{k}{m x_0^4} T_0^2} .$$

Thus $T_0 = x_0^2 \sqrt{m/k}$.

2. (20pts.) A particle moves in the one-dimensional half-space $(x > 0)$. The potential energy function is

$$U(x) = -\frac{k}{x^3} + \frac{l}{x^2} \quad , \quad (x > 0)$$

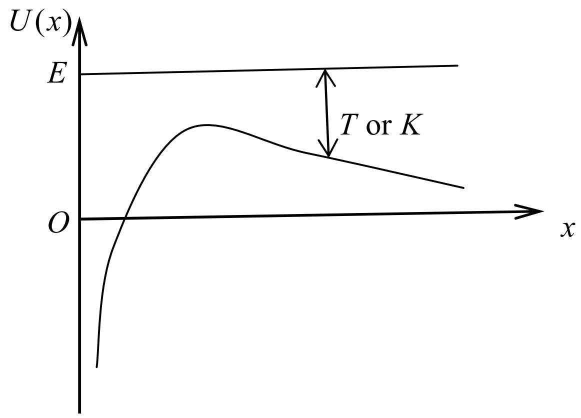
where k and l are both positive constants.

(a) Sketch the potential energy function. If the mechanical energy of the particle is $E\,(> 0)$, indicate the kinetic energy of the particle at a position in the potential energy diagram.

(b) Find the corresponding force field acting on the particle.

(c) Determine the finite equilibrium position of the particle and identify it as stable or unstable.

Solution: (a) As shown in the figure drawn.



(b) The corresponding force field is

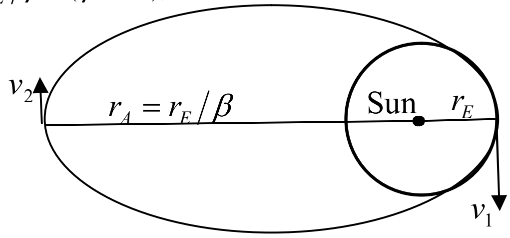
$$F(x) = -\frac{dU}{dx} = -\frac{3k}{x^4} + \frac{2l}{x^3}, \, (x > 0).$$

(c) $F(x) = -\frac{3k}{x^4} + \frac{2l}{x^3} = 0 \Rightarrow x_{\mathrm{eq}} = \frac{3k}{2l}$.

$$\left.\frac{d^2U}{dx^2}\right|_{x=x_{\mathrm{eq}}} = \left.\left(-\frac{12k}{x^5} + \frac{6l}{x^4}\right)\right|_{x=x_{\mathrm{eq}}} = -\frac{32}{81}\frac{l^5}{k^4} < 0 .$$

The equilibrium point is unstable.

3. (20pts.) Disposal of nuclear waste poses a serious environmental threat for human being. It is proposed that the nuclear waste can be dumped on the sun by using the delivery rocket. One of the processes is that the delivery rocket will first be launched to escape the earth's gravity and to travel at earth's orbital speed v_E around the sun. And the retrorockets will be fired to bring the speed of the delivery rocket to zero and it will then fall straight into the sun. Here the orbit of the earth around the sun can be approximately taken as a circle of radius r_E. However, this process is very fuel consuming. A much more economical way is to boost the rocket in the same direction of the earth's orbital motion to a speed v_1 $(> v_E)$ so that it will follow an elongated elliptical orbit with the sun as one of its fuci. At the aphelion of the orbit where the furthest distance of the rocket from the sun is $r_A = r_E/\beta$ $(\beta < 1)$, the rocket has much lower speed v_2 and is much easier to bring to the rest by retrorockets. The mass of the sun is denoted as M_{sun}.



(a) Find orbital speed of the earth v_E in terms of r_E and M_{sun}.

(b) Show that $v_1^2 = 2v_E^2/(1+\beta)$.

(c) Show that $v_2 = \beta v_1 = \sqrt{2}\beta v_E / \sqrt{1+\beta}$.

Solution: (a) The equation of motion (uniform circular motion) is

$$m\frac{v_E^2}{r_E} = G\frac{M_{\text{sun}} m}{r_E^2} .$$

After some algebra, we obtain

$$v_E^2 = \frac{GM_{\text{sun}}}{r_E} \quad \text{or} \quad v_E = \sqrt{\frac{GM_{\text{sun}}}{r_E}} .$$

(b) and (c) The conservation of the angular momentum gives

$$m v_1 r_E = m v_2 r_A . \tag{1}$$

The conservation of energy gives

$$\frac{1}{2}mv_1^2 - \frac{GM_{\text{sun}}m}{r_E} = \frac{1}{2}mv_2^2 - \frac{GM_{\text{sun}}m}{r_A}. \quad (2)$$

Substituting $r_A = r_E/\beta$ into Eq. (1), we have

$$mv_1 r_E = mv_2 \frac{r_E}{\beta}, \qquad v_2 = \beta v_1 \quad (3)$$

Substituting $r_A = r_E/\beta$ and $v_2 = \beta v_1$ into Eq. (2), we have

$$\frac{1}{2}mv_1^2 - \frac{GM_{\text{sun}}m}{r_E} = \frac{1}{2}m\beta^2 v_1^2 - \beta\frac{GM_{\text{sun}}m}{r_E}.$$

Thus

$$(1-\beta^2)v_1^2 = 2(1-\beta)\frac{GM_{\text{sun}}}{r_E} \quad \text{or} \quad v_1^2 = \frac{2GM_{\text{sun}}}{(1+\beta)r_E} = \frac{2v_E^2}{1+\beta},$$

and

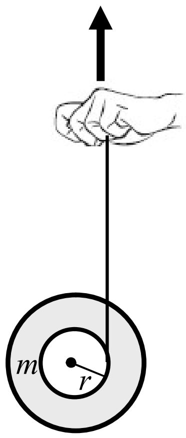
$$v_2 = \beta v_1 = \frac{\sqrt{2}\beta v_E}{\sqrt{1+\beta}}.$$

4. (20pts.) A child plays a toy yoyo that consists of two disks concentrically connected by an axel of radius r. The mass of the yoyo is m and its rotational inertia around the axel is $I = \gamma m r^2$ where γ is a positive constant. A thread is wrapped around the axel and the mass of the thread is negligible. When the yoyo is released from rest, the child pulls the free end of the thread upward so that the position of the center of mass of the yoyo remains unchanged.

(a) Find the tensional force of the thread.

(b) Find the angular acceleration of the yoyo.

(c) Find the acceleration of the hand of the child.



Solution: (a) The equation of motion of the center of mass is

$$T - mg = ma_c.$$

Since $a_c = 0$, $T = mg$.

(b) The equation of the rotational motion of the yoyo is

$$I\alpha = Tr.$$

Thus

$$\alpha = \frac{Tr}{I} = \frac{mgr}{\gamma m r^2} = \frac{g}{\gamma r}.$$

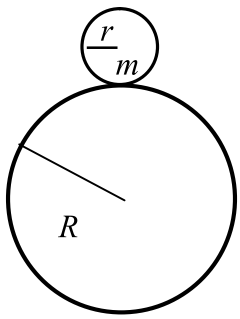
(c) The acceleration of the child's hand is

$$a_h = \alpha r = \frac{g}{\gamma}.$$

5. (20 pts.) A small uniform cylinder of mass m and radius r rests initially on the top of the outer surface of a fixed cylinder of radius R. With a tiny kick, the small cylinder rolls down without slipping.

(a) Find the position where the small cylinder breaks off from the outer surface of the cylinder.

(b) Find the angular velocity of the small cylinder when it flies off from the outer surface of the cylinder.

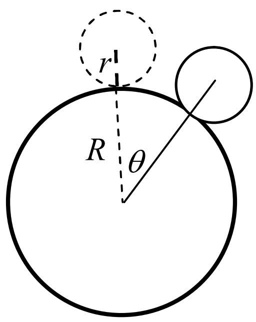


Solution:

(a) The conservation of mechanical energy gives

$$mg(R+r)(1-\cos\theta) = \frac{1}{2}mv_C^2 + \frac{1}{2}\left(\frac{1}{2}mr^2\right)\omega^2.$$

The condition for rolling without slipping is $v_C = \omega r$. Substituting it into above equation, we have



$$mg(R+r)(1-\cos\theta)=\frac{3}{4}mv_C^2 \quad \text{or} \quad m\frac{v_C^2}{R+r}==\frac{4}{3}mg(1-\cos\theta). \qquad (1)$$

The radial equation of motion of the center of mass is

$$m\frac{v_C^2}{R+r}=mg\cos\theta-N. \qquad (2)$$

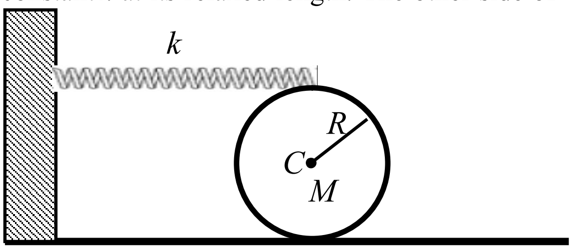
And the condition for the break-off is $N=0$. Substituting $N=0$ and Eq. (1) into Eq. (2), we have

$$\frac{4}{3}mg(1-\cos\theta)=mg\cos\theta \quad \Rightarrow \quad \cos\theta=\frac{4}{7}, \quad \theta=\arccos\frac{4}{7}.$$

(b) Substituting $\cos\theta=\dfrac{4}{7}$ into Eq. (1), we obtain

$$v_C^2=\frac{4}{7}g(R+r), \ v_C=\sqrt{\frac{4g(R+r)}{7}}, \quad \omega=\frac{v_C}{r}=\sqrt{\frac{4g(R+r)}{7r^2}}.$$

6. (20pts.) As shown in the accompanying figure, a solid cylinder of radius R and mass M is on a horizontal surface. The top of the cylinder is connected to a horizontal spring with spring constant k at its relaxed length. The other side of the spring is attached to a wall. The cylinder can roll without slipping on the surface.



(a) If the cylinder rolls to the right so that its center of mass (COM) makes a displacement x_C, what is the stretch of the spring?

(b) Write down the equation of translational motion of the COM of the cylinder and the equation of rotational motion of the cylinder.

(c) What is the angular frequency of small oscillations of the cylinder?

Solution: (a) Since the cylinder rolls without slipping on the surface, when its center of mass displaces a distance x_C, it also rotates an angle $\alpha=x_C/R$. Thus, the spring stretches out a length $x_C+\alpha R=2x_C$.

(b) The equation of translational motion is

$$M\frac{d^2x_C}{dt^2}=-k(2x_C)+F_f. \qquad (1)$$

The equation of rotational motion is

$$\frac{1}{2}MR^2\frac{d^2\alpha}{dt^2}=-k(2x_C)R-F_fR. \qquad (2)$$

The condition for rolling without slipping is $x_C=\alpha R$.

(c) Substituting $x_C=\alpha R$ ($\alpha=x_C/R$) into Eq.(2), we obtain

$$\frac{1}{2}M\frac{d^2x_C}{dt^2}=-k(2x_C)-F_f \qquad (3)$$

(1)+(3) $\Rightarrow$:

$$\frac{3}{2}M\frac{d^2x_C}{dt^2}=-4kx_C \quad \text{or} \quad \frac{d^2x_C}{dt^2}+\frac{8k}{3M}x_C=0.$$

Thus $\omega^2=\dfrac{8k}{3M}$, $\omega=\sqrt{\dfrac{8k}{3M}}$.